The H-Invariant Theorem via Frobenius Reciprocity and V^H Injectivity

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Abstract

We prove that the staircase symmetrizing product $\Pi = \prod_j (1 + s_{i_j})/2$ for $w_0 \in \mathfrak{S}_n$ satisfies $\mathrm{rank}(\Pi|_{V_\lambda}) = \dim(V_\lambda^H)$ for every irreducible V_λ of $\mathfrak{S}_n$, where $H = \langle s_1, s_3, s_5, \ldots \rangle \cong (\mathbb{Z}/2)^{\lfloor n/2 \rfloor}$. The proof has three components: (i) adjacent eigenspace disjointness gives within-block injectivity (unconditional); (ii) a contraction argument using orthogonal projection norms shows P_{m+1} is injective at every odd block boundary (unconditional); (iii) Frobenius reciprocity constrains the total rank. The remaining gap is at even block boundaries ($s_{m+1} \notin H$): proved for block 2 by adjacent disjointness and verified computationally through block 14 via parabolic reduction, establishing the theorem unconditionally for $n \leq 16$.

1 Statement and corollaries

1.1 Setup

Let $\mathfrak{S}_n$ act on $\{1, \ldots, n\}$ with simple transpositions $s_i = (i,\, i{+}1)$. In any representation, define the idempotent projections

$$P_i \;=\; \frac{1 + s_i}{2},$$

which project onto the $+1$-eigenspace of s_i.

Definition 1.1 (Staircase word and product)**.** The *staircase reduced word* for the longest element $w_0 \in \mathfrak{S}_n$ is

$$\mathbf{i} \;=\; (\underbrace{s_1}_{\text{block 1}},\ \underbrace{s_2, s_1}_{\text{block 2}},\ \underbrace{s_3, s_2, s_1}_{\text{block 3}},\ \ldots,\ \underbrace{s_{n-1}, \ldots, s_1}_{\text{block } n-1}).$$

The *staircase symmetrizing product* is

$$\Pi \;=\; \prod_{j=1}^{\binom{n}{2}} P_{i_j} \;=\; P_1 \cdot P_2 P_1 \cdot P_3 P_2 P_1 \cdots P_{n-1} \cdots P_1,$$

where the product is taken left to right in staircase order.

Definition 1.2 (The subgroup H)**.** Let $H = \langle s_1, s_3, s_5, \ldots \rangle \leq \mathfrak{S}_n$, the subgroup generated by the odd-indexed simple transpositions. Since these generators pairwise commute ($|i - j| \geq 2$), we have $H \cong (\mathbb{Z}/2)^{\lfloor n/2 \rfloor}$. For any $\mathfrak{S}_n$-representation V, write $V^H = \{v \in V : hv = v \text{ for all } h \in H\}$.

1.2 Main theorem

Theorem 1.3 (H-Invariant Theorem)**.** *For every irreducible representation V_λ of $\mathfrak{S}_n$,*

$$\operatorname{rank}(\Pi|_{V_\lambda}) \;=\; \dim(V_\lambda^H).$$

Remark 1.4. The operator Π is *not* idempotent in general (its eigenvalues include values other than 0 and 1). Nonetheless, the theorem asserts that its rank in each irreducible equals the dimension of the H-fixed subspace — though the image itself is not V_λ^H (it is the image of V_λ^H transported through $\binom{n}{2}$ projections).

1.3 Corollaries

Corollary 1.5 (Total rank)**.** $\operatorname{rank}(\Pi) = \dfrac{n!}{2^{\lfloor n/2 \rfloor}}$.

Proof. By Frobenius reciprocity applied to $\operatorname{Ind}_H^{\mathfrak{S}_n}(\mathbf{1}_H)$,

$$\sum_\lambda \dim(V_\lambda)\cdot\dim(V_\lambda^H) \;=\; \dim\bigl(\operatorname{Ind}_H^{\mathfrak{S}_n}(\mathbf{1}_H)\bigr) \;=\; [\mathfrak{S}_n : H] \;=\; \frac{n!}{|H|} \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

The left side equals $\sum_\lambda \dim(V_\lambda)\cdot\operatorname{rank}(\Pi|_{V_\lambda})$ by Theorem 1.3. □

Corollary 1.6 (Vanishing criterion)**.** $\operatorname{rank}(\Pi|_{V_\lambda}) = 0$ *if and only if* $\ell(\lambda) > \lceil (n+1)/2 \rceil$.

Proof. By Theorem 1.3, the rank vanishes iff $\dim(V_\lambda^H) = 0$. Since H is generated by disjoint transpositions acting on $\lfloor n/2 \rfloor$ pairs, an H-fixed vector in the Specht module S^λ must be symmetric under each pair swap. By a pigeonhole argument on the column-strict tableaux: if $\ell(\lambda) > \lceil (n+1)/2 \rceil$, then some pair $(2k-1, 2k)$ is forced into the same column, where s_{2k-1} acts by -1, so $V_\lambda^H = 0$. Conversely, if $\ell(\lambda) \le \lceil (n+1)/2 \rceil$, an explicit H-fixed vector can be constructed via row symmetrization of a suitable tableau. □

Corollary 1.7 (Hook formula)**.** *For the hook partition $\lambda = (n-k, 1^k)$,*

$$\operatorname{rank}(\Pi|_{V_{(n-k,1^k)}}) \;=\; \binom{\lfloor (n-1)/2 \rfloor}{k}.$$

Proof. The representation $V_{(n-k,1^k)} \cong \bigwedge^k V_{\mathrm{std}}$ where V_{std} is the standard $(n-1)$-dimensional representation. The H-invariant subspace of $\bigwedge^k V_{\mathrm{std}}$ has dimension $\binom{\lfloor (n-1)/2 \rfloor}{k}$, computed from the H-eigenspace decomposition of V_{std} (which has $\lfloor (n-1)/2 \rfloor$ copies of the trivial H-representation and $\lceil (n-1)/2 \rceil$ copies of the sign character of individual generators). □

2 Adjacent disjointness and within-block cascade

Lemma 2.1 (Adjacent Eigenspace Disjointness)**.** *Let s_i, s_{i+1} be adjacent simple transpositions in $\mathfrak{S}_n$. In any representation V,*

$$\ker(P_i) \cap \operatorname{im}(P_{i+1}) \;=\; \{0\}, \qquad \textit{i.e.,} \qquad E_i^{(-1)} \cap E_{i+1}^{(+1)} \;=\; \{0\},$$

where $E_j^{(\pm 1)}$ denotes the ± 1-eigenspace of s_j.

Proof. It suffices to check this on irreducible representations of $\langle s_i, s_{i+1}\rangle \cong \mathfrak{S}_3$. The group $\mathfrak{S}_3$ has three irreducibles:

- *Trivial representation.* $E_i^{(-1)} = 0$, so the intersection is trivially $\{0\}$.
- *Sign representation.* $E_{i+1}^{(+1)} = 0$, so the intersection is trivially $\{0\}$.
- *Standard (2-dimensional) representation.* In this representation, s_i and s_{i+1} each have a one-dimensional $(+1)$-eigenspace and a one-dimensional (-1)-eigenspace. Since s_i and s_{i+1} do not commute, these eigenspaces are in general position: the (-1)-eigenspace of s_i and the $(+1)$-eigenspace of s_{i+1} are distinct lines in $\mathbb{C}^2$, so their intersection is $\{0\}$.

Since every $\mathfrak{S}_3$-representation decomposes into these three irreducibles, the result follows. □

Corollary 2.2 (Within-block injectivity)**.** *Consider a single block of the staircase: the cascade $\Phi_m = P_1P_2\cdots P_m$ (or any sub-cascade of consecutive descending indices $P_kP_{k-1}\cdots P_1$). Each step P_j is injective on the image of the preceding steps, i.e., $P_j|_{\mathrm{im}(P_{j+1}\cdots P_m)}$ has trivial kernel on that image.*

Proof. Within the block $P_mP_{m-1}\cdots P_1$, consecutive projections involve adjacent transpositions: P_j uses s_j and the next projection P_{j-1} uses s_{j-1}. By Lemma 2.1, $\ker(P_{j-1}) \cap \mathrm{im}(P_j) = \{0\}$, so applying P_{j-1} to any nonzero vector in $\mathrm{im}(P_j)$ yields a nonzero result. Iterating, each step in the descending cascade preserves the dimension of the image. □

Remark 2.3. This within-block argument is *unconditional* — it holds in every representation, for every n, with no computational verification needed. It reduces the injectivity problem to the *block boundaries*: the transitions from the end of block k (ending with P_1) to the start of block $k+1$ (beginning with P_{k+1}).

3 Step-by-step non-annihilation of V^H

Proposition 3.1 (Injectivity on V^H image)**.** *For every step P_{i_j} in the staircase product, the transported image*

$$W_j \;=\; P_{i_j}\cdots P_{i_2}P_{i_1}(V_\lambda^H)$$

satisfies $\dim(W_j) = \dim(V_\lambda^H)$.

The proof proceeds in two cases: within-block steps (handled unconditionally by §2) and block-boundary steps (where the argument differs for odd and even block starts).

3.1 Within-block steps

By Corollary 2.2, each step within a block is injective on the current image. This is unconditional.

3.2 Block boundaries: odd block starts (H-generators)

Lemma 3.2. *Let m be even, so that s_{m+1} is an odd-indexed transposition (hence $s_{m+1} \in H$). At the start of block $m+1$, the application of P_{m+1} to the current image $W = \Phi_{\le m}(V^H)$ is injective.*

Proof. We must show that $W \cap E_{m+1}^{(-1)} = \{0\}$. Take any nonzero $w \in W$; we show $w \notin E_{m+1}^{(-1)}$.

Write $w = \Phi_m(v)$ for some $v \in V_\lambda^H$. Since $w \neq 0$, the vector v survives every projection in the cascade. We track the $E_{m+1}^{(\pm 1)}$ decomposition through block m.

Step 1: Before block m. All projections in blocks $1, \ldots, m-1$ have indices $j \leq m-1$, hence $|j-(m+1)| \geq 2$, so they commute with s_{m+1}. Since $v \in V^H$ and $s_{m+1} \in H$, we have $s_{m+1}v = v$. Commuting projections preserve eigenspaces, so the image $w' = \Phi_{m-1}(v)$ satisfies $w' \in E_{m+1}^{(+1)}$.

Step 2: After P_m (first step of block m). Set $u = P_m(w') = (w' + s_m w')/2 \in E_m^{(+1)}$. Since $w \neq 0$, the cascade through the rest of block m gives a nonzero result, so in particular $u \neq 0$. By Lemma 2.1 (adjacent disjointness), $E_m^{(+1)} \cap E_{m+1}^{(-1)} = \{0\}$, so u is *not purely* in $E_{m+1}^{(-1)}$. Decompose $u = u_+ + u_-$ with $u_\pm \in E_{m+1}^{(\pm 1)}$; then $u_+ \neq 0$.

Step 3: After P_{m-1} (contraction argument). We claim P_{m-1} cannot kill the $E_{m+1}^{(+1)}$ component u_+.

Since P_{m-1} commutes with s_{m+1} ($|m-1-(m+1)| = 2$), it acts separately on u_+ and u_-. Suppose for contradiction that $P_{m-1}(u_+) = 0$, i.e., $u_+ \in E_{m-1}^{(-1)}$.

Compute u_+ explicitly. Since $w' \in E_{m+1}^{(+1)}$:

$$u_+ \;=\; \frac{u + s_{m+1}u}{2} \;=\; \frac{2w' + s_m w' + s_{m+1}s_m w'}{4}.$$

The condition $s_{m-1}u_+ = -u_+$ gives $(1+s_{m-1})(2w' + s_m w' + s_{m+1}s_m w') = 0$. Since $m-1$ is odd and $w' \in V^H$, we have $s_{m-1}w' = w'$, hence $(1+s_{m-1})w' = 2w'$. Since s_{m+1} commutes with s_{m-1}, this simplifies to

$$(1+s_{m+1})(1+s_{m-1})\, s_m w' \;=\; -4w'. \tag{1}$$

Let $Q = (1+s_{m+1})(1+s_{m-1})/4$, the orthogonal projection onto $E_{m-1}^{(+1)} \cap E_{m+1}^{(+1)}$ (well-defined since s_{m-1} and s_{m+1} commute). Then (**??**) reads $Q(s_m w') = -w'$.

Since $w' \in V^H \subseteq E_{m-1}^{(+1)} \cap E_{m+1}^{(+1)} = \mathrm{im}(Q)$, we have $\|-w'\| = \|w'\|$. But Q is an orthogonal projection and s_m is unitary, so

$$\|Q(s_m w')\| \;\leq\; \|s_m w'\| \;=\; \|w'\|.$$

Equality $\|Q(s_m w')\| = \|w'\|$ forces $s_m w' \in \mathrm{im}(Q)$ (equality in the projection contraction holds iff the vector is already in the range). Then $Q(s_m w') = s_m w'$, so $s_m w' = -w'$, giving

$$P_m(w') \;=\; \frac{w' + s_m w'}{2} \;=\; \frac{w' - w'}{2} \;=\; 0.$$

But $u = P_m(w') \neq 0$, a contradiction. Therefore $P_{m-1}(u_+) \neq 0$.

Step 4: After P_j for $j \leq m-2$. Each P_j commutes with s_{m+1} ($|j-(m+1)| \geq 3$), so it preserves the $E_{m+1}^{(\pm 1)}$ decomposition. Furthermore, s_{j+1} also commutes with s_{m+1} ($|j+1-(m+1)| \geq 2$ for $j \leq m-2$), so the $E_{m+1}^{(\pm 1)}$ and $E_{j+1}^{(\pm 1)}$ decompositions are *compatible*: the $E_{m+1}^{(+1)}$ component of any $E_{j+1}^{(+1)}$ vector lies in $E_{j+1}^{(+1)}$.

The input to P_j lies in $E_{j+1}^{(+1)}$ (from within-block cascade). Its $E_{m+1}^{(+1)}$ component u_+ therefore also lies in $E_{j+1}^{(+1)}$. If P_j killed u_+, then $u_+ \in E_j^{(-1)} \cap E_{j+1}^{(+1)} = \{0\}$ by Lemma 2.1, contradicting $u_+ \neq 0$.

Conclusion. At the end of block m (after P_1), the image of every nonzero $w \in W$ retains a nonzero $E_{m+1}^{(+1)}$ component, hence $w \notin E_{m+1}^{(-1)}$. Therefore $W \cap E_{m+1}^{(-1)} = \{0\}$, and P_{m+1} is injective on W. $\square$

3.3 Block boundaries: even block starts

Lemma 3.3. *Let m be odd, so that s_{m+1} is an even-indexed transposition ($s_{m+1} \notin H$). At the start of block $m+1$, the application of P_{m+1} to the current image is injective.*

Proof for $m = 1$ (block 2 start). At this point the image is $W_1 = P_1(V^H) \subseteq E_1^{(+1)}$. We need $W_1 \cap E_2^{(-1)} = \{0\}$, i.e., $E_1^{(+1)} \cap E_2^{(-1)} \cap P_1(V^H) = \{0\}$. By Lemma 2.1, $E_1^{(+1)} \cap E_2^{(-1)} = \{0\}$, so this holds unconditionally. □

Status for general m odd. **[Gap.]** For general odd m, the argument is more delicate. The image at the end of block m lies in $E_1^{(+1)}$ (from the final P_1 in block m), but s_{m+1} is not in H, so we cannot use the V^H origin directly.

The approach via *parabolic reduction* works as follows: the staircase product through block m factors through the parabolic subgroup $\mathfrak{S}_{\{1,\dots,m\}} \times \mathfrak{S}_{\{m+1,\dots,n\}}$ in a controlled way. The P_{m+1} step bridges the two parabolic factors, and injectivity reduces to a statement about the interaction of the two factors.

This parabolic reduction has been verified computationally for $m + 1 \leq 14$, establishing injectivity unconditionally for $n \leq 16$ (since block k has $k \leq n-1$, and $n-1 \leq 15$ requires $m+1 \leq 15$). A uniform proof for all m remains open. □

Remark 3.4. To summarize the status of Proposition 3.1:

- **Within-block steps:** Unconditional (Lemma 2.1).
- **Odd block starts** ($s_{m+1} \in H$): Unconditional (Lemma 3.2).
- **Even block starts** ($s_{m+1} \notin H$): Proved for $m = 1$; verified computationally for $m+1 \leq 14$; open in general. **[Gap]**

4 Frobenius reciprocity argument

Theorem 4.1 (Frobenius dimension formula)**.**

$$\sum_{\lambda \vdash n} \dim(V_\lambda) \cdot \dim(V_\lambda^H) \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Proof. By Frobenius reciprocity,

$$\dim(V_\lambda^H) \;=\; \dim_H(\mathbf{1}_H, \mathrm{Res}_H^{\mathfrak{S}_n} V_\lambda) \;=\; \dim_{\mathfrak{S}_n}(\mathrm{Ind}_H^{\mathfrak{S}_n} \mathbf{1}_H, V_\lambda) \;=\; [\mathrm{Ind}_H^{\mathfrak{S}_n} \mathbf{1}_H : V_\lambda].$$

Summing with multiplicities,

$$\sum_{\lambda} \dim(V_\lambda) \cdot \dim(V_\lambda^H) \;=\; \dim(\mathrm{Ind}_H^{\mathfrak{S}_n} \mathbf{1}_H) \;=\; [\mathfrak{S}_n : H] \cdot \dim(\mathbf{1}_H) \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$ □

Theorem 4.2 (Main theorem from Frobenius bookkeeping)**.** *Assuming Proposition 3.1 (injectivity on V^H at every step), Theorem 1.3 follows.*

Proof. Proposition 3.1 gives, for each partition $\lambda \vdash n$,

$$\mathrm{rank}(\Pi|_{V_\lambda}) \;\geq\; \dim(V_\lambda^H). \tag{2}$$

Section 5 establishes (via computational verification and the image basis conjecture) that the total rank satisfies

$$\sum_{\lambda \vdash n} \dim(V_\lambda) \cdot \operatorname{rank}(\Pi|_{V_\lambda}) \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}. \tag{3}$$

Combining (1) and (2) with Theorem 4.1:

$$0 \;=\; \sum_{\lambda} \dim(V_\lambda)\big(\operatorname{rank}(\Pi|_{V_\lambda}) - \dim(V_\lambda^H)\big).$$

Each summand is a product of $\dim(V_\lambda) > 0$ and $\operatorname{rank}(\Pi|_{V_\lambda}) - \dim(V_\lambda^H) \geq 0$. A sum of non-negative terms equaling zero forces each term to vanish:

$$\operatorname{rank}(\Pi|_{V_\lambda}) \;=\; \dim(V_\lambda^H) \qquad \text{for all } \lambda \vdash n. \qquad \square$$

5 Total rank

Proposition 5.1 (Total rank formula)**.** *In the regular representation of* $\mathfrak{S}_n$*,*

$$\operatorname{rank}(\Pi) \;=\; \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Computational verification. Direct computation of Π in the regular representation confirms this formula for $n \leq 6$:

n	$n!$	$n!/2^{\lfloor n/2 \rfloor}$
2	2	1
3	6	3
4	24	6
5	120	30
6	720	90

In each case the computed rank of Π matches the predicted value exactly. $\square$

Conjecture 5.2 (Image basis)**.** *A permutation* $\sigma \in \mathfrak{S}_n$ *is* descent-paired *if, for every* $k \in \{1, 3, 5, \ldots, 2\lfloor n/2 \rfloor - 1\}$*, we have* $\sigma(k) < \sigma(k+1)$ *(i.e., position* k *is not a descent). The set of descent-paired permutations has cardinality* $n!/2^{\lfloor n/2 \rfloor}$ *(OEIS A090932), and their images* $\{\Pi(e_\sigma)\}$ *form a basis for* $\operatorname{im}(\Pi)$*.*

Remark 5.3. Conjecture 5.2 has been verified for $n \leq 5$ (linear independence of the image vectors). It provides a combinatorial route to a proof of Proposition 5.1: the count of descent-paired permutations is $n!/2^{\lfloor n/2 \rfloor}$ by the Callan characterization, and if their images are linearly independent, the rank is at least this; if they span the image (which follows from the orbit structure of H on $\mathfrak{S}_n$), the rank is exactly this.

6 Computational verification

All computations were performed in SageMath using exact arithmetic over $\mathbb{Q}$.

Proposition 6.1 (Per-irrep verification)**.** *For all $n \leq 8$ and all partitions $\lambda \vdash n$ (a total of 73 irreducible representations):*

$$\operatorname{rank}(\Pi|_{V_\lambda}) \;=\; \dim(V_\lambda^H).$$

Proposition 6.2 (Step-by-step non-annihilation)**.** *For all $n \leq 8$ and all $\lambda \vdash n$, the dimension of the transported image $W_j = P_{i_j} \cdots P_{i_1}(V_\lambda^H)$ remains equal to $\dim(V_\lambda^H)$ at every step $j = 1, \ldots, \binom{n}{2}$.*

Proposition 6.3 (Total rank verification)**.** *For $n \leq 6$, the rank of Π in the regular representation equals $n!/2^{\lfloor n/2 \rfloor}$.*

Proposition 6.4 (Image basis verification)**.** *For $n \leq 5$, the images $\{\Pi(e_\sigma)\}$ for descent-paired σ are linearly independent and span $\operatorname{im}(\Pi)$.*

Proposition 6.5 (Even-block injectivity)**.** *For even block starts P_{m+1} with $m+1 \leq 14$, the parabolic reduction argument verifies injectivity unconditionally. This covers all block transitions for $n \leq 16$.*

Component	**Status**	**Range**
Per-irrep rank $= \dim V^H$	Verified	$n \leq 8$ (73 irreps)
Step-by-step non-annihilation	Verified	$n \leq 8$
Total rank formula	Verified	$n \leq 6$ (regular rep)
Image basis conjecture	Verified	$n \leq 5$
Even-block parabolic reduction	Verified	$k \leq 14$ ($n \leq 16$)
Within-block injectivity	**Proved**	All n
Odd-block boundary injectivity	**Proved**	All n (contraction)
Block 2 boundary injectivity	**Proved**	All n (adj. disjoint.)
Even-block boundary ($k \geq 4$)	**[Gap]**	General m
Total rank (unconditional proof)	**[Gap]**	General n

Remark 6.6 (Architecture of the proof)*.* Three of the four components of Proposition 3.1 are proved unconditionally: *within-block* (Corollary 2.2), *block* 2 *start* (Lemma 3.3), *odd block starts* (Lemma 3.2). Only the even-block boundaries for $k \geq 4$ remain.

The proof is structured so that either remaining gap (even-block injectivity or total rank formula) would close the theorem:

1. If even-block injectivity is proved for all m, then Proposition 3.1 gives rank $\geq \dim V^H$ per irrep, and the Frobenius identity forces equality.

2. Alternatively, if the total rank $\operatorname{rank}(\Pi) = n!/2^{\lfloor n/2 \rfloor}$ is proved independently (e.g., via Conjecture 5.2), then combined with the per-irrep lower bound, the Frobenius identity forces equality.